Project Proposal

SWE 4740

Embedded Software Development Lab

Autonomous Smart Parking & Revenue Management System

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1 Introduction

With the rapid growth of vehicles in urban areas, efficient parking management has become a significant challenge. Traditional parking systems rely heavily on manual supervision, which often leads to congestion, time wastage, and security risks. An automated solution is necessary to improve parking efficiency, ensure security, and reduce human dependency. This project proposes an Autonomous Smart Parking Management System with Security and Billing, which integrates sensing, control, and monitoring mechanisms into a single embedded system. Using simulation platforms such as Tinkercad and Proteus Design Suite, the system demonstrates how real-time slot detection, automated entry control, and time-based billing can be implemented to create a smarter and more secure parking environment.

2 Problem Statement

Manual parking systems often operate without automation, making them inefficient and difficult to manage in busy environments. Drivers usually struggle to find available parking spaces, which results in unnecessary delays and contributes to traffic congestion. These systems also lack proper monitoring and security mechanisms, allowing unauthorized access and making parking management less reliable. Furthermore, traditional setups do not include automated time tracking or billing, requiring manual supervision and increasing the chances of errors. To overcome these limitations, an Autonomous embedded solution is needed to ensure efficient space usage, secure access control, and automated parking management.

Key Problems:

- Inefficient utilization of parking spaces
- Difficulty in identifying available parking slots
- Increased traffic congestion due to searching for parking
- Lack of real-time monitoring
- Absence of secure access control
- Risk of unauthorized vehicle entry
- No automated parking time tracking
- Manual and error-prone billing process

3 Project Idea

To address the inefficiencies and security limitations of traditional parking systems described in the problem statement, this project proposes the development of an Autonomous Smart Parking Management System that automates the core parking operations using embedded system

technology. The primary objective of the system is to minimize human involvement by enabling automatic detection, controlled entry, real-time monitoring, and time-based billing within a simulated smart parking environment.

The system operates by continuously monitoring the occupancy status of each parking slot using ultrasonic sensors. These sensors detect the presence or absence of vehicles and provide real-time feedback to the control unit. Based on this information, LED indicators are used to visually represent slot availability — green indicating an empty slot and red indicating an occupied slot. This allows both users and administrators to easily understand the parking status without manual inspection.

To regulate vehicle access, an automated entry gate mechanism is implemented using a servo motor. The gate opens only when two conditions are satisfied: a parking slot is available and the user is successfully authenticated through a keypad-based security system. This ensures that unauthorized vehicles cannot enter the parking facility, thereby improving overall system security.

Once a vehicle enters the parking area, the system begins tracking the parking duration using embedded timing logic. When the vehicle exits, the total parking time is calculated and used to determine the corresponding parking fee based on predefined billing rules. This automated billing process eliminates manual record keeping and reduces the possibility of human error.

All relevant system information, including total parking slots, available spaces, assigned slot number, and calculated parking fee, is displayed in real time on a 16×2 LCD module. This enhances transparency and usability for both operators and users.

The complete system is implemented and tested through simulation using platforms such as Tinkercad or Proteus Design Suite. By integrating sensing, control, security, and billing functionalities, the proposed system demonstrates a practical and scalable embedded solution for modern automated parking management.

4 System Features

The proposed Autonomous Smart Parking Management System integrates multiple functional modules to provide a fully automated parking solution. Each feature of the system works in coordination to ensure efficient space utilization, secure access, real-time monitoring, and accurate billing. From detecting available parking slots to controlling vehicle entry, tracking parking duration, calculating fees, and providing system status via an LCD display, the embedded system is designed to handle all critical aspects of modern parking management. The following section details each feature and its operation within the system.

4.1 Parking Slot Monitoring

The system continuously monitors each parking slot to detect vehicle presence, ensuring efficient space utilization.

- **Ultrasonic Sensors:** Detect the presence or absence of a vehicle in each slot. Measures distance to determine whether a slot is occupied.
- **LED Indicators:**
  - **Green LED:** Indicates the slot is empty and available for parking.
  - **Red LED:** Indicates the slot is occupied, preventing new entries into that space.
- **Real-Time Updates:** Slot availability is immediately updated to the control unit for display on the LCD, ensuring accurate information for both users and administrators.

4.2 Automatic Entry Gate

The system controls vehicle entry using an automated servo-controlled gate to enhance security and traffic flow.

- **Servo Motor:** Acts as a physical barrier that opens or closes based on system conditions.
- **Conditional Access:** Gate opens only when:
  1. At least one parking slot is available.
  2. User is successfully authenticated through the security system.
- **Failsafe Logic:** If the slot is unavailable or authentication fails, the gate remains closed, preventing unauthorized or excess entry.
- **Smooth Operation:** Gate movement is controlled via PWM signals from the microcontroller for precise opening and closing.

4.3 Parking Time Calculation

The system tracks the time each vehicle spends in the parking facility to enable accurate billing.

- **Start Timer:** Begins as soon as a vehicle enters the parking lot after authorization.
- **Stop Timer:** Stops automatically when the vehicle exits.
- **Time Logging:** Duration is stored in memory for fee calculation and potential reporting.
- **Accuracy:** Uses microcontroller timing functions to ensure precise tracking, minimizing billing errors.

4.4 Billing System

Automatically calculates the parking fee based on duration and predefined rates.

- **Rate Structure:**
  1. First 1 hour = 50 Taka
  2. Every additional hour = 10 Taka
- **Automated Calculation:** Billing is performed by the microcontroller based on entry and exit time.
- **Display:** Fee is displayed on the LCD in real time for transparency.
- **Error Reduction:** Eliminates manual calculation, preventing overcharging or undercharging.

4.5 Security System

Ensures that only authorized vehicles can access the parking facility, improving overall safety.

- **Authentication Method:** Keypad-based password entry or simulated RFID logic.
- **Failed Attempts Handling:** After 3 incorrect attempts, a buzzer is triggered to alert administrators.
- **Access Control:** Denies entry automatically if authentication fails, preventing unauthorized usage.
- **Audit Trail:** System logs failed attempts, which can be used for monitoring security incidents.

4.6 LCD Display

Provides a user-friendly interface to display all relevant parking information.

- **Information Displayed:**
  1. Total number of parking slots
  2. Number of available slots
  3. Assigned parking slot for each vehicle
  4. Calculated parking fee
  5. System messages such as “Gate Open,” “Full Parking,” or “Invalid Password”
- **Real-Time Updates:** Ensures that both drivers and administrators receive immediate feedback on parking status.
- **Enhanced Usability:** Helps users quickly identify available spaces and understand system status without manual assistance.

5 Required Elements

Feature	Required Components	Purpose / Role
Parking Slot Monitoring	Ultrasonic Sensors	Detect presence of vehicles in each slot
	Green & Red LEDs	Visual indication of slot availability
Automatic Entry Gate	Servo Motor	Controls gate opening and closing
Parking Time Calculation	Arduino Internal Timer	Tracks entry and exit duration accurately
Billing System	LCD Display	Shows calculated parking fee and slot info
Security System	Keypad / RFID Module	Authenticates users before entry
	Buzzer	Alerts in case of unauthorized access attempts
Entry/Exit Control	Push Buttons	Simulates vehicle entry and exit events
Overall Control	Arduino Microcontroller	Coordinates sensors, actuators, and display

Table 1: System Features, Components, and Roles

6 Challenges We May Face

Developing the Autonomous Smart Parking Management System involves several technical and operational challenges that need to be addressed for accurate and reliable functionality:

- **Sensor Calibration and Accuracy:** Ultrasonic sensors must be properly calibrated to accurately detect vehicle presence without false positives or negatives. Variations in vehicle size and placement may affect readings.
- **Timing and Synchronization:** Entry and exit events must be precisely timed to ensure accurate calculation of parking duration and billing. Delays in processing sensor input could lead to incorrect fee calculation.
- **Gate Control Reliability:** The servo motor controlling the entry gate must operate smoothly and respond in real-time. Mechanical delays or incorrect PWM signals could result in gate malfunction.
- **Security Management:** Implementing a secure authentication system via keypad or RFID simulation requires careful logic to prevent unauthorized access while avoiding accidental

lockouts.

- **Multi-Vehicle Handling:** The system must handle multiple vehicles entering or exiting simultaneously without errors in slot assignment or fee calculation.
- **Real-Time Display Updates:** The LCD must update parking slot availability, assigned slots, and fees instantly, which requires efficient coding and management of data flow.
- **Integration of Components:** Coordinating multiple hardware components (sensors, LEDs, servo, buzzer, LCD) with the microcontroller requires careful design to prevent signal interference or timing conflicts.
- **Simulation Limitations:** When using Tinkercad or Proteus for simulation, some hardware behaviors may differ from real-life implementation, which could require adjustments in the actual deployment.

7 Future Work and Improvements

The proposed system is designed to be modular and scalable, allowing for enhancements and advanced features in the future:

- **Over-Capacity Logic:**
  - When parking slots reach full capacity, the keypad input will be temporarily locked.
  - The LCD will display a scrolling "PARKING FULL" message to inform users and prevent unauthorized entries.
- **Dynamic Pricing:**
  - Introduce algorithmic pricing based on occupancy levels.
  - Example: Increase parking rates during peak hours when occupancy exceeds 80%, promoting optimal usage of space and generating additional revenue.
- **Power Saving / Sleep Mode:**
  - Implement low-power mode during idle hours when no vehicles are present.
  - The system can be awakened by interrupts triggered by vehicle detection or button presses, reducing energy consumption and enhancing sustainability.
- **Diagnostic Serial Monitor:**
  - Add a hidden "Technician Mode" that outputs system health data via serial communication to a PC.
  - This will include details such as total vehicles parked, daily revenue, sensor status, and error logs, enabling efficient maintenance and troubleshooting.
- **Multi-Level Parking Expansion:**

  - The system can be upgraded to manage multiple floors with slot allocation for each level, using additional microcontrollers and communication protocols.

- **Integration with Mobile Application:**
  - Future integration with a mobile app could allow users to check available slots remotely, reserve slots, and make digital payments for parking fees.